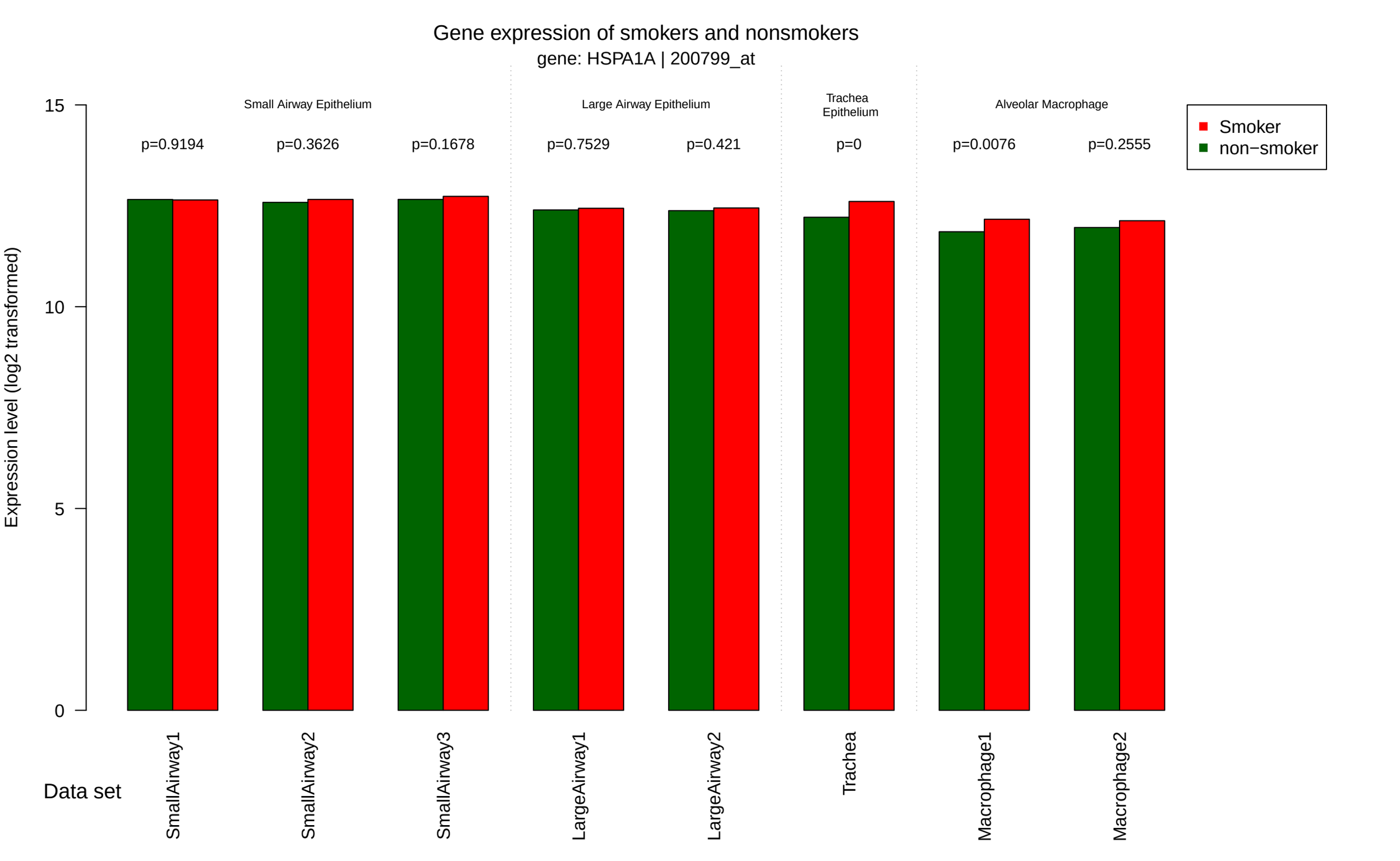
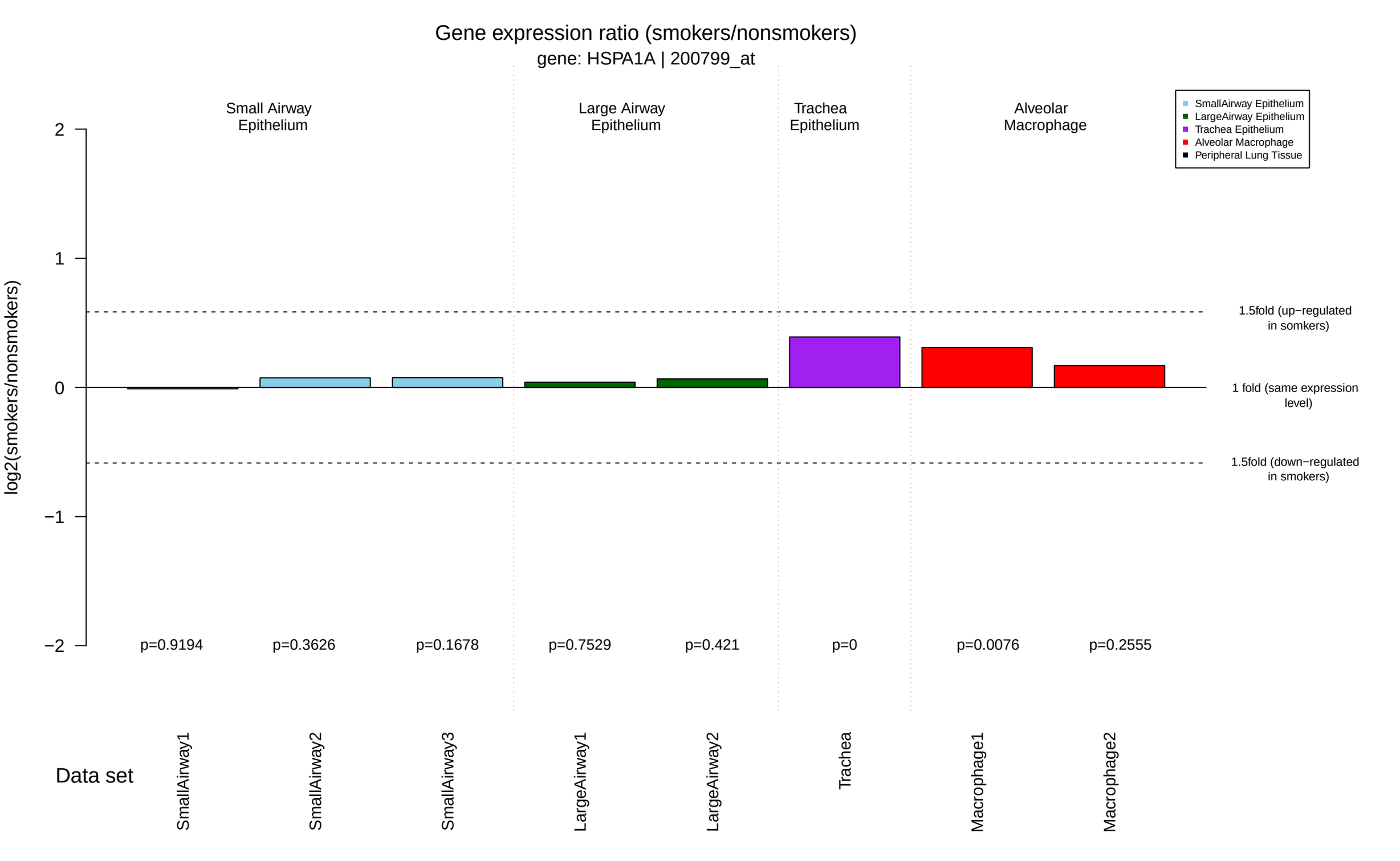
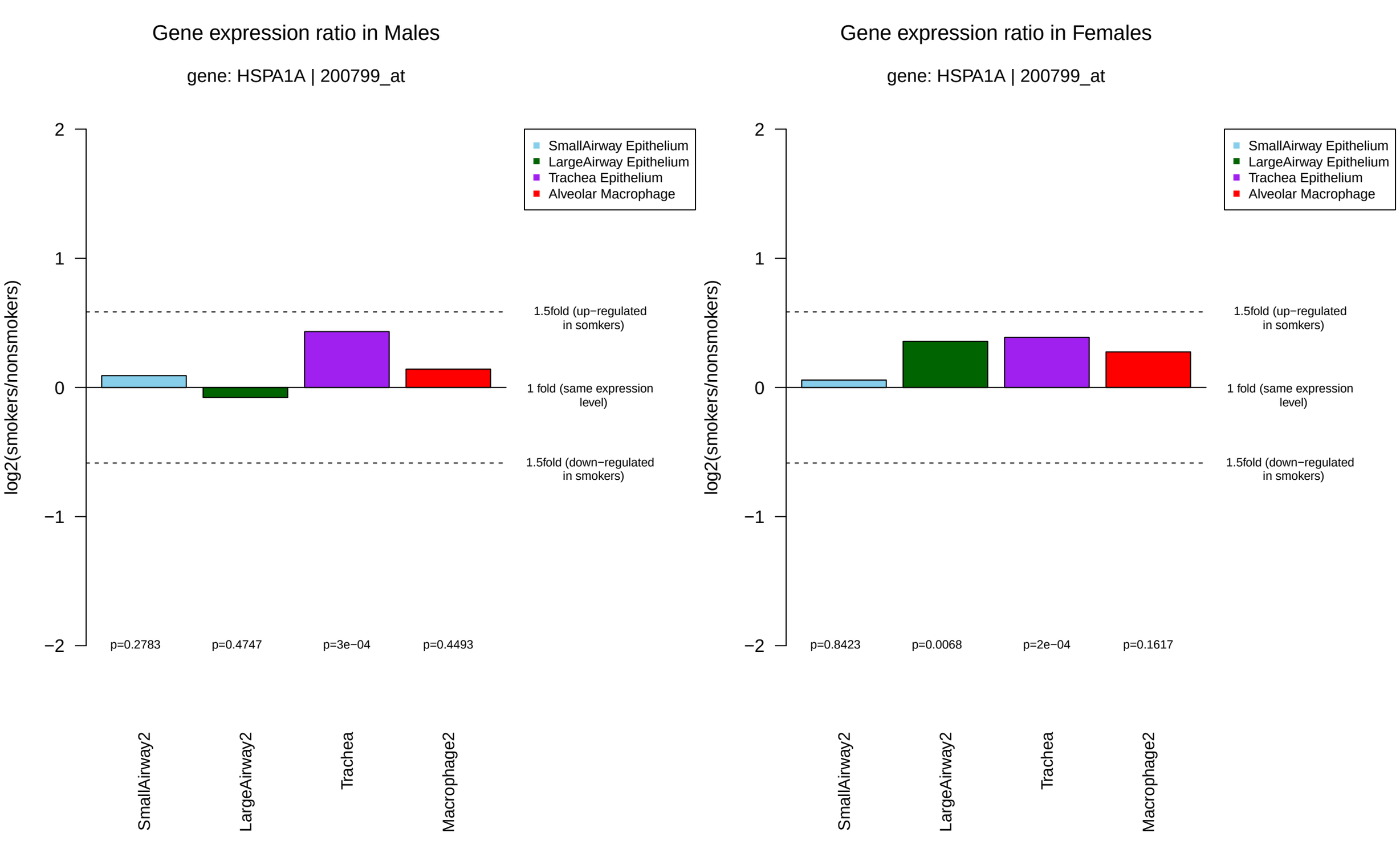
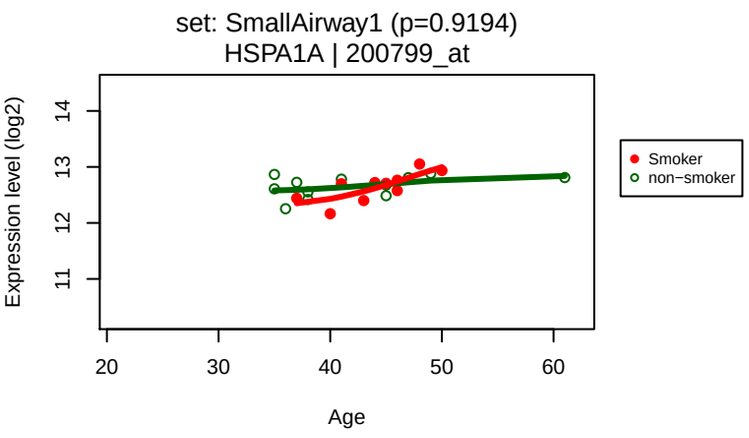
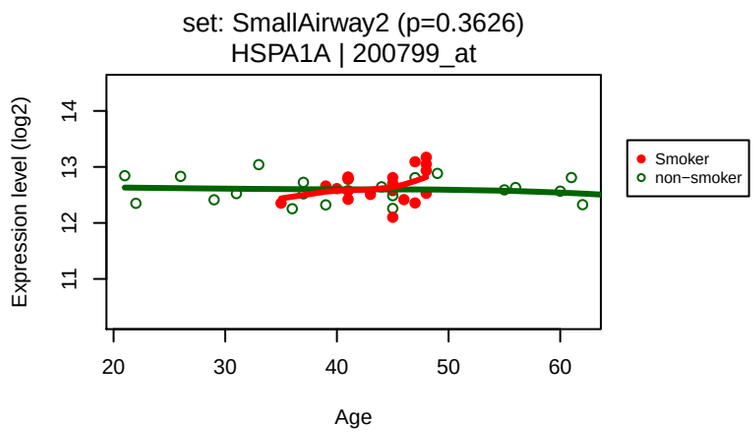
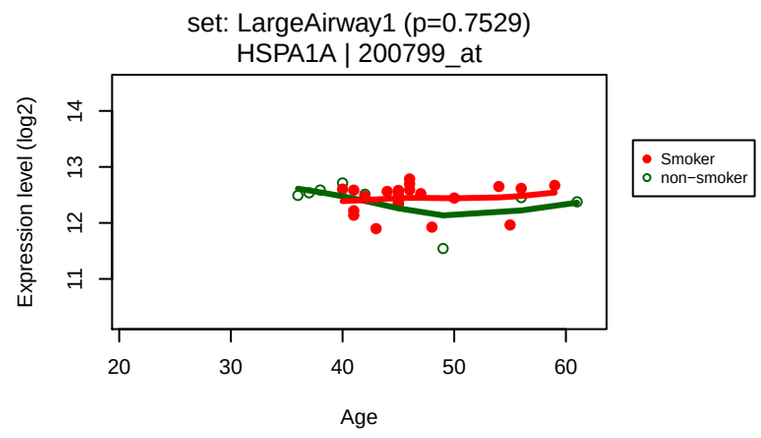

Gene expression of smokers and nonsmokers
gene: HSPA1A | 200799_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9194
p=0.3626
p=0.1678
p=0.7529
p=0.421
p=0
p=0.0076
p=0.2555
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

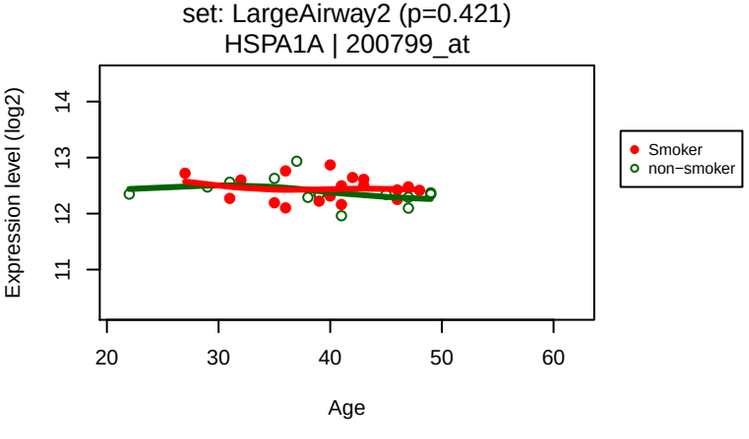

Gene expression ratio (smokers/nonsmokers)
gene: HSPA1A | 200799_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9194
p=0.3626
p=0.1678
p=0.7529
p=0.421
p=0
p=0.0076
p=0.2555
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

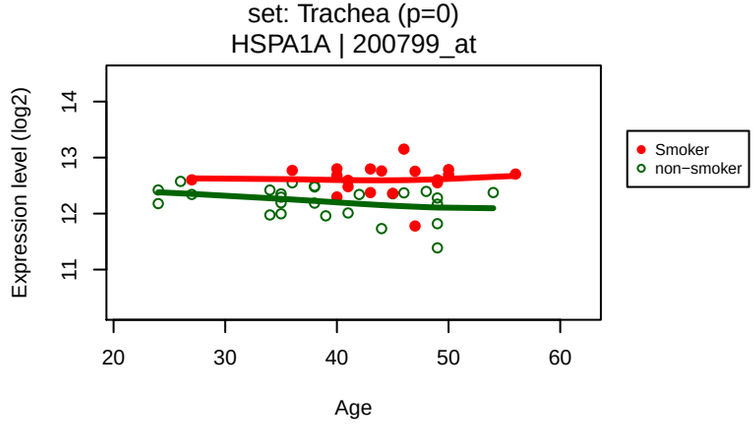

Gene expression ratio in Males
gene: HSPA1A | 200799_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2783
p=0.4747
p=3e−04
p=0.4493
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA1A | 200799_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8423
p=0.0068
p=2e−04
p=0.1617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

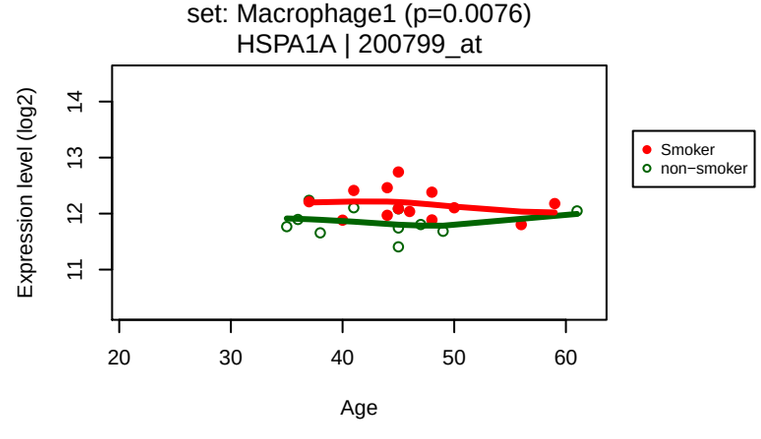

set: SmallAirway1 (p=0.9194)
HSPA1A | 200799_at
Expression level (log2)
Age
Smoker
non−smoker

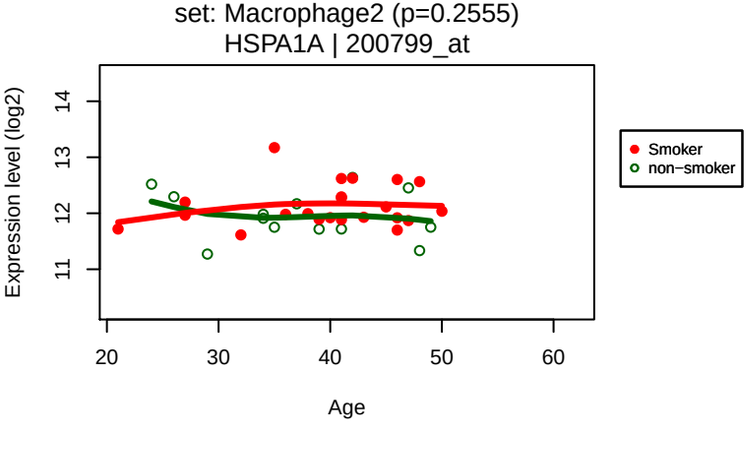

set: SmallAirway2 (p=0.3626)
HSPA1A | 200799_at
Expression level (log2)
Age
Smoker
non−smoker

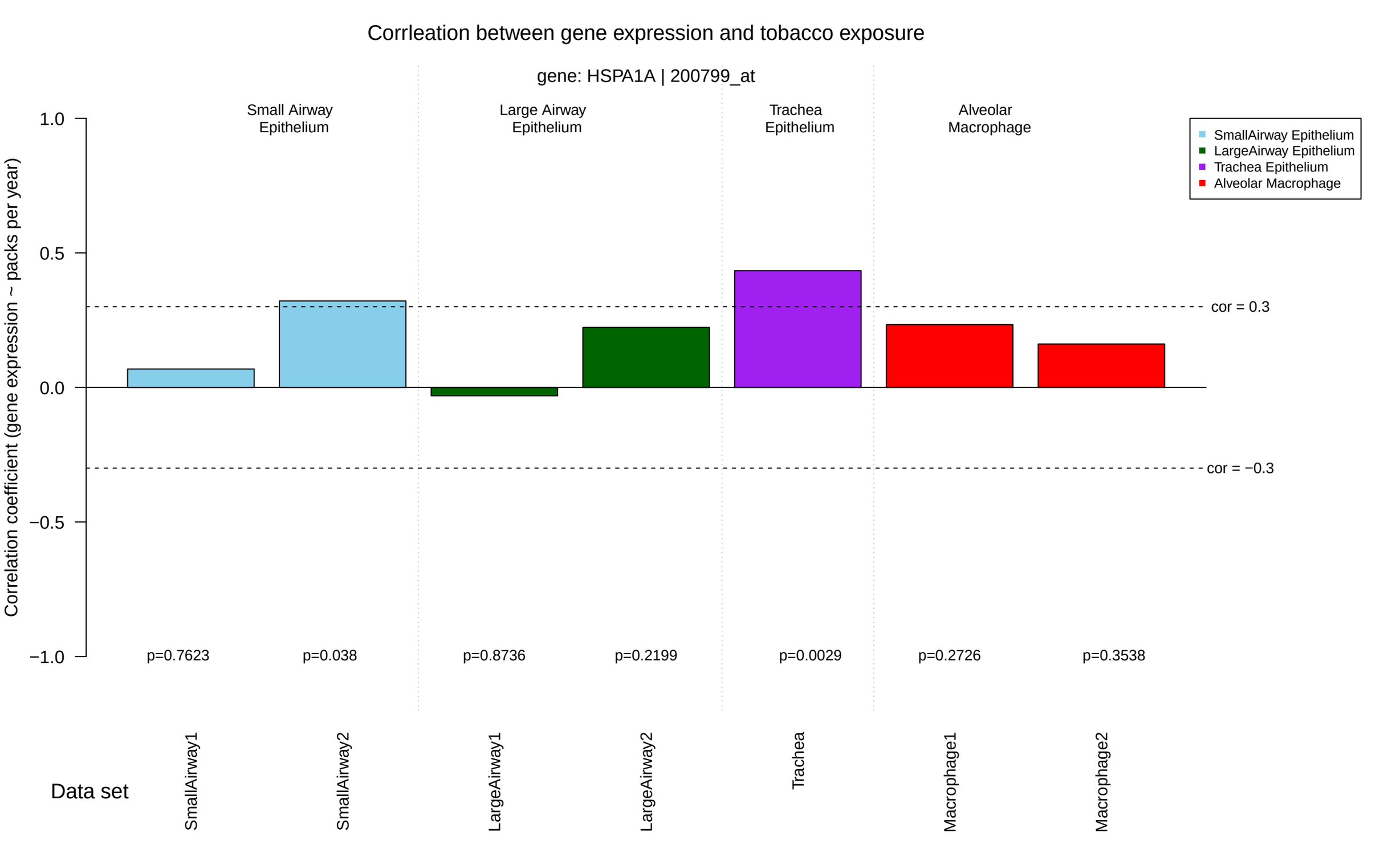

set: LargeAirway1 (p=0.7529)
HSPA1A | 200799_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.421)
HSPA1A | 200799_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
HSPA1A | 200799_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0076)
HSPA1A | 200799_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2555)
HSPA1A | 200799_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA1A | 200799_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7623
p=0.038
p=0.8736
p=0.2199
p=0.0029
p=0.2726
p=0.3538
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2